

Do LLM fingerprints survive outside English?

Interpretable attribution of five frontier models across seven languages, under translation, and against the models as judges

Adrian Erlikhman¹ Michael Tarekegn¹ Philo Juang²

1 Los Angeles Center for Enriched Studies (LACES) 2 University of California, Los Angeles

2026 IEEE MIT Undergraduate Research Technology Conference, Technology of Computation



code, prompts, all data, full report
github.com/adrian-erlikhman/LangLLM

TL;DR

Twenty-one interpretable features defined on Universal Dependencies attribute text to one of five frontier LLMs at 59% to 72% accuracy in all seven languages (chance 20%), and accuracy does not fall from English to Hindi. The fingerprint is largely language-invariant, survives Google Translate and an LLM translator, and is invisible to the models themselves, which judge authorship at chance. Models do converge stylistically as resources fall, so the register of low-resource languages is flattening.

1 Motivation

Detection of AI-written text is mature and is known to weaken outside English. Attributing a text to a **specific** model is a different task: it has been studied almost only in English, with opaque embedding classifiers.

Yet the need for attribution, in disinformation tracing, academic integrity and moderation, is mostly non-English. If fingerprints fade with a language's training-data share, attribution fails where it matters most. If models converge on one style in low-resource languages, LLM use is flattening the written register of the languages with the least data.

No prior work had measured attribution with interpretable features across a resource gradient, separated model style from language style, tested stylistic convergence, or checked whether translation removes the signal.

2 Study design

Models	GPT-5.5, Gemini 3.5 Flash, Claude Opus 4.7, Grok 4.3, DeepSeek V4 Pro, all through one OpenRouter key with served model strings logged
Languages	English, Spanish, Chinese, Russian, Japanese, Turkish, Hindi, in decreasing training-data share (rank 1 to 7)
Prompts	12 persuasive-essay prompts; each fixes a topic, a stance (6 for, 6 against) and three supporting points, so content is held constant and only style varies
Generation	single user turn, no system prompt, temperature 0.7, fixed seeds, hidden reasoning excluded

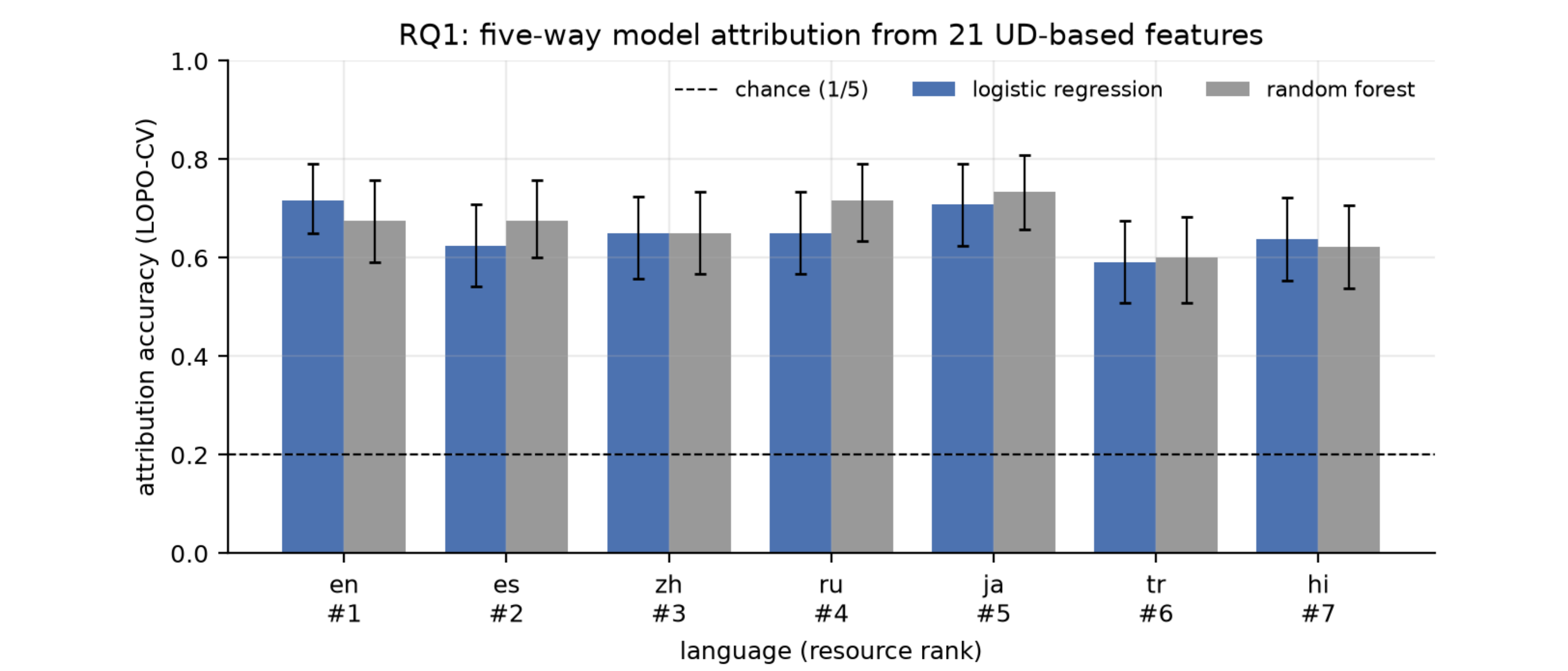


3 Features

Every text is parsed with Stanza on Universal Dependencies, so each feature means the same thing in Chinese, Turkish and Hindi. English-only measures (Flesch, Fog, hedges, passive voice, contractions) are excluded.

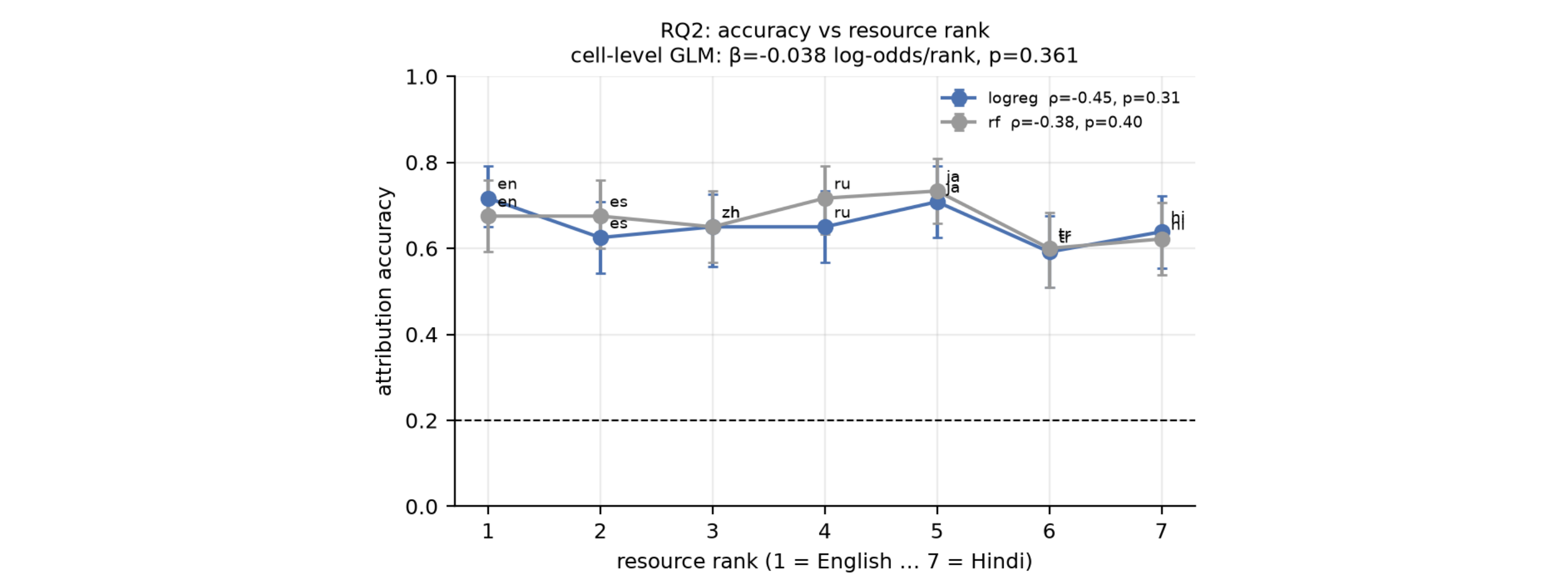
Lexical	moving-average type-token ratio, hapax rate, mean token length, Zipf slope
Syntactic	sentence length mean and SD, burstiness, dependency depth, subordinate-clause rate, function-word ratio, first-person rate
Structure	paragraph count and length, question rate, connective rate
Punctuation	comma, colon, dash, semicolon per 1,000 tokens, script equivalents mapped
Character	character-bigram entropy, digit rate

4 Attribution works in every language



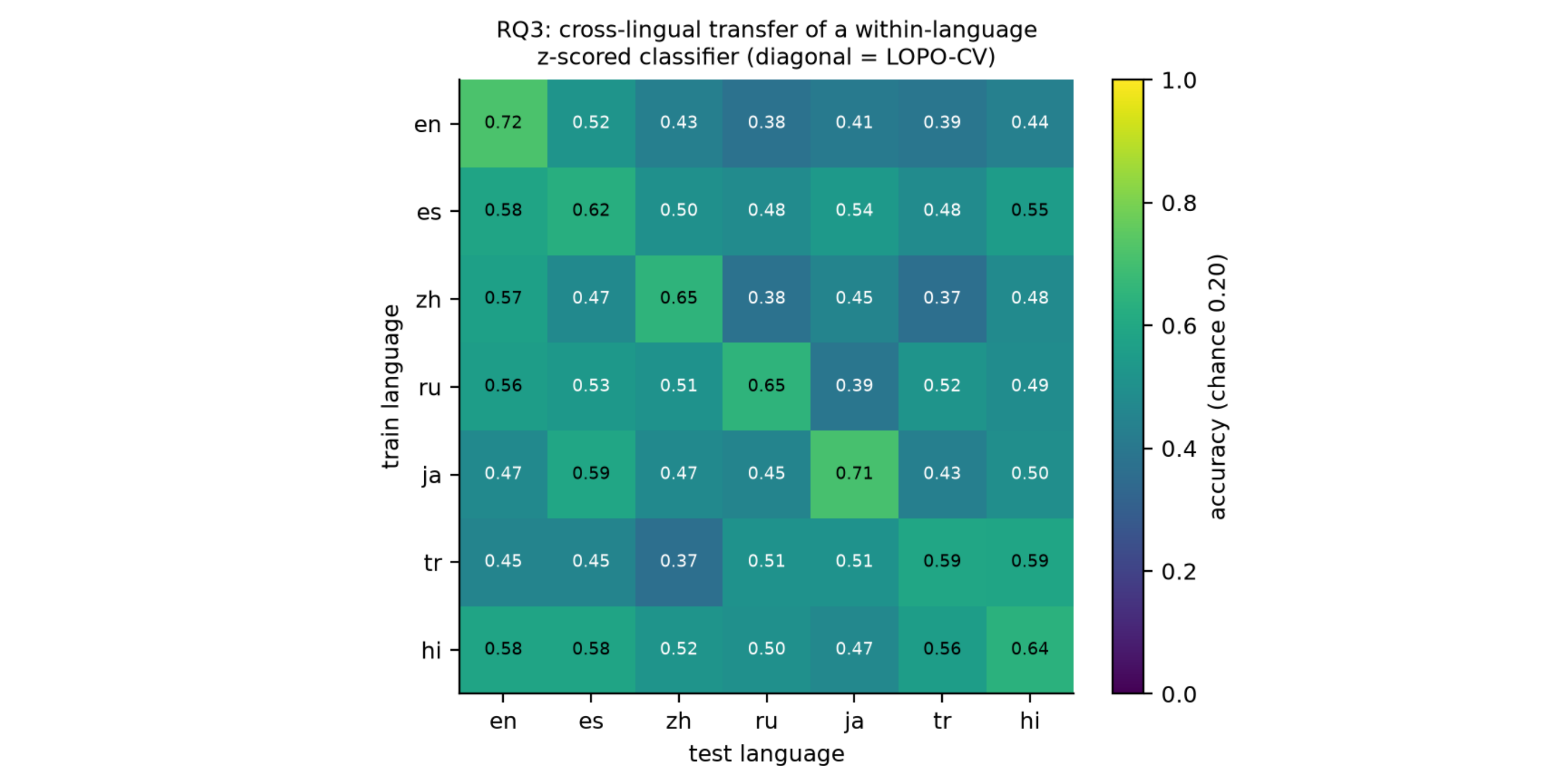
Five-way accuracy per language, leave-one-prompt-out cross-validation (both generations of a prompt stay on one side). Logistic regression on standardised features is the interpretable primary; a random forest is the ceiling. Every 95% interval excludes chance. GPT-5.5 and Grok 4.3 are the easiest to identify, DeepSeek the blurriest. The signal is structural: paragraph count, comma and colon density and lexical-diversity slope carry most of it; dependency depth and subordination contribute little.

5 No resource gradient



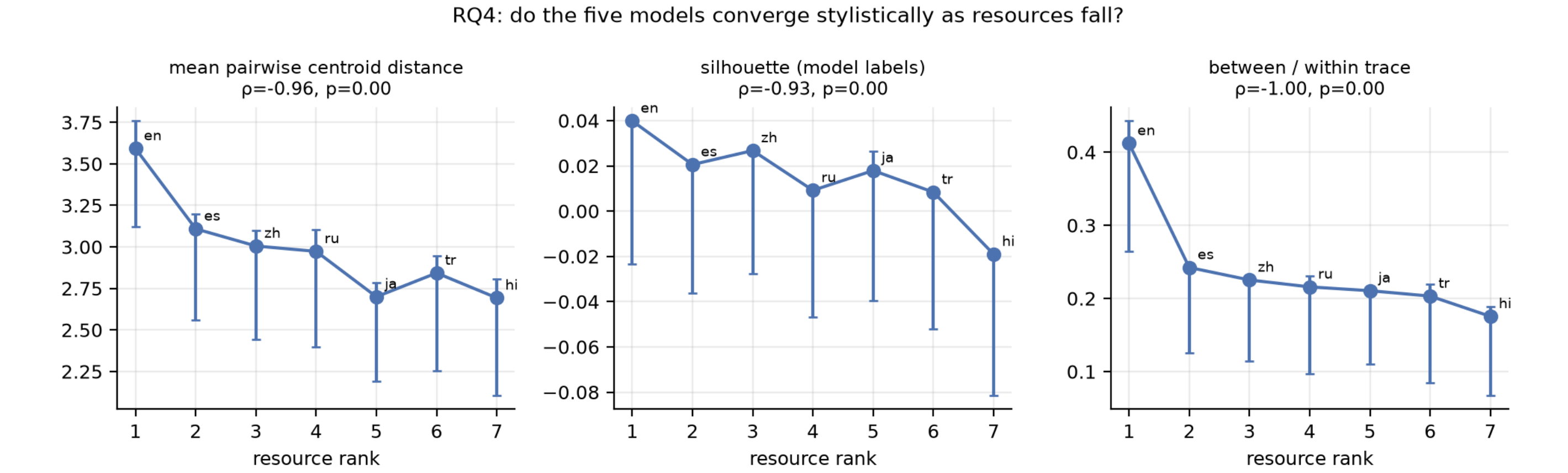
Spearman rho = -0.45 (p = 0.31) across the seven languages; a cell-level binomial GLM with prompt-clustered errors gives -0.038 log-odds per rank step (p = 0.36, n = 839). Japanese matches English and Hindi exceeds Spanish. The pre-registered gradient hypothesis is not supported.

6 Model style is mostly language-invariant



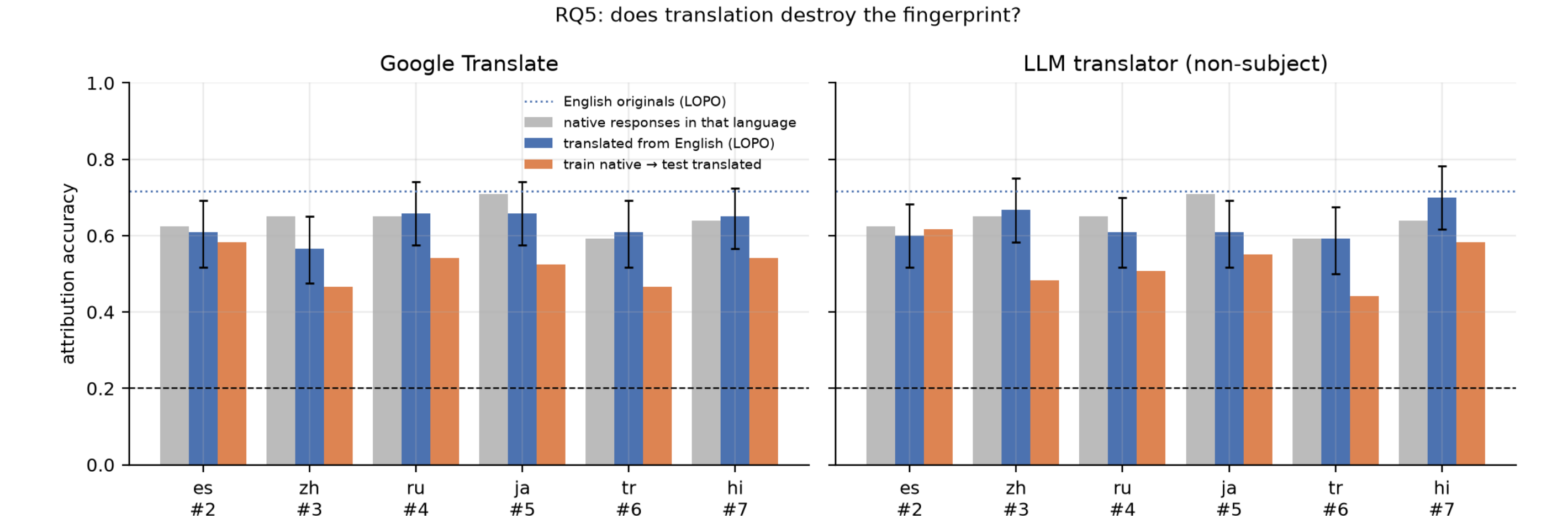
Language explains most feature variance (mean partial eta squared 0.54 for language, 0.14 for model, 0.13 for their interaction). Remove each language's mean, and a classifier trained on any one language beats chance on every other: 42 of 42 pairs, mean 0.49 against 0.65 within language.

7 Models converge as resources fall



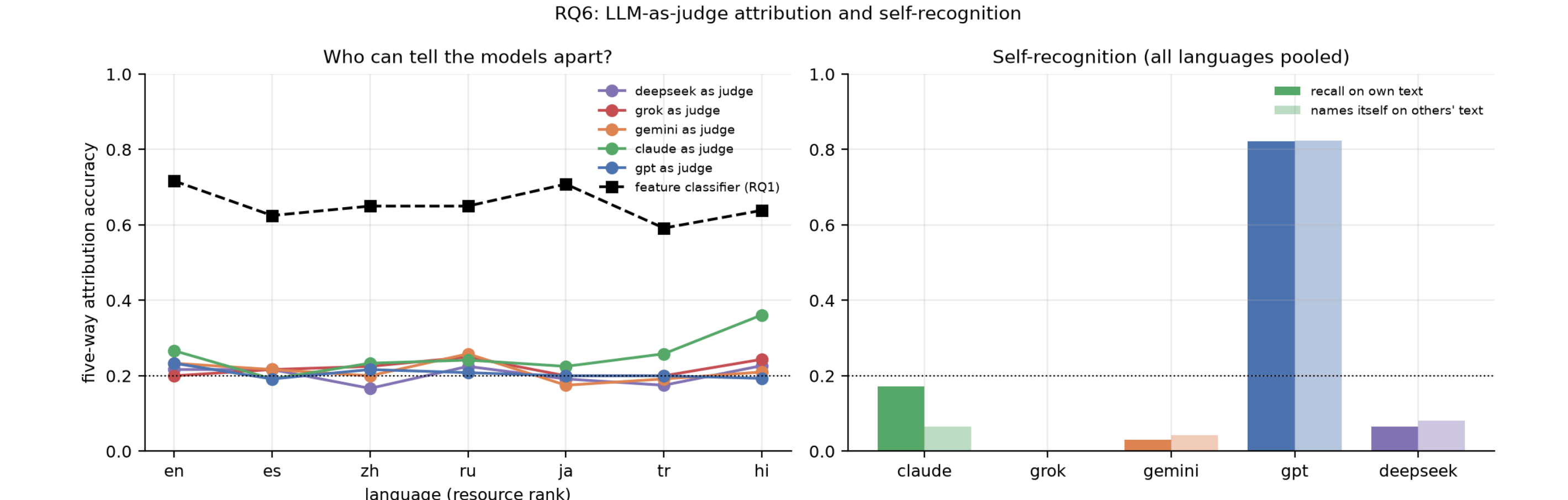
Within-language separation of the five model centroids falls monotonically from English to Hindi (centroid distance rho = -0.96; between-to-within ratio 0.41 in English, 0.18 in Hindi). Almost all of the change is Grok's terse English register disappearing into the pack. Convergence in the bulk of features; attribution sustained by a few.

8 Translation does not remove the fingerprint



All 120 English responses were translated into the six other languages by Google Translate and by a free open model (MiniMax M3). Attribution on translated text alone is 0.62 and 0.63; a classifier trained on the English originals reads the translations at 0.66. Structure survives translation (paragraph count rho = 1.00, sentence length 0.89); vocabulary is rewritten (MATTR about 0.3).

9 Baseline: the models themselves cannot do this



Each model was shown each essay and asked which of the five wrote it (five names in shuffled order, temperature 0, hidden reasoning off). GPT-5.5 20.6%, Grok 21.9%, Gemini 21.2%, DeepSeek 20.3%; chance in all seven languages, each defaulting to one fixed answer. Claude, at 25.4%, is the only judge above chance, and only just.

Takeaways

Interpretable, cross-lingually defined features attribute LLM text in every language tested and beat every LLM judge by 35 to 50 points. The fingerprint does not fade with resource level and survives translation, so provenance across a language boundary works from an English-trained classifier. The five models write more alike in Hindi than in English: the register of low-resource languages is flattening. Limits: one genre; length is part of the linear signal; ordinal resource rank; dated model snapshots.